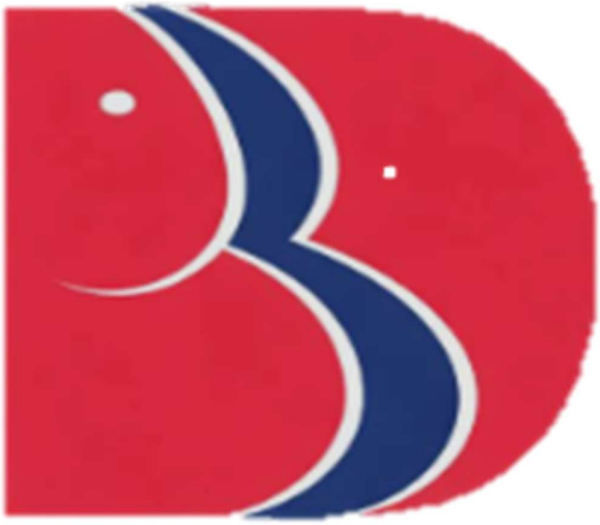


BBD UNIVERSITY

2026-2027

SCHOOL OF COMPUTER APPLICTION

No SQL PROJECT:

Book Inventory Management System

Submitted to:

Mr. Harendra Singh

Department: BCA(DS & AI

Date: 28th September

Submitted by:

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INTRODUCTION

BookVault – Book Inventory Management System is a project developed using MongoDB, a NoSQL database management system. The main objective of this project is to store, manage, search, and organize book information efficiently. This project contains a collection named books with 30 book records, including details such as book ID, title, author, category, price, publication year, and rating.

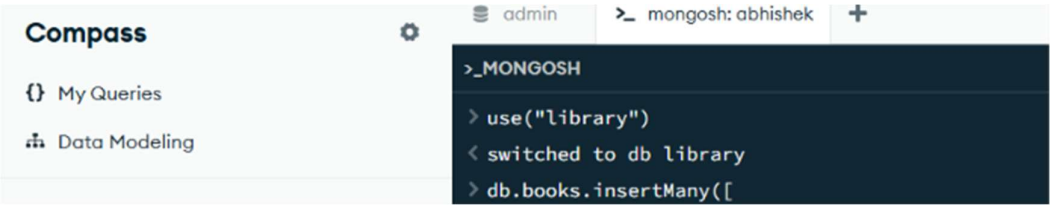
The project demonstrates various MongoDB operations, including inserting, retrieving, filtering, sorting, updating, and counting book records. It also uses projection and aggregation to display specific information and summarize books by category. Through this project, we can understand the practical implementation of MongoDB queries and learn how NoSQL databases manage structured information efficiently.

1.CREATE DATABASE

MongoDB Query:

Use library

Explanation: Creates or switches to the library database.

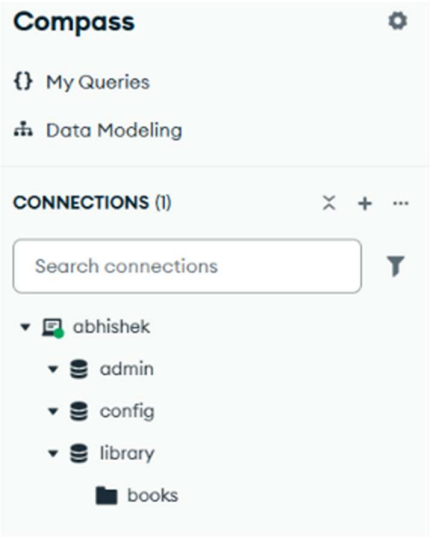


2. CREATE COLLECTION

MongoDB Query:

db. create Collection(“books”)

Explanation: Creates a collection named books.



3. INSERT BOOKS DETAILS

MongoDB Query: db.books.insertMany([

```
{
  bookId: 1,
  title: "The Alchemist",
  author: "Paulo Coelho",
  category: "Fiction",
  price: 350,
  year: 1988,
  rating: 8.0
},
{
  bookId: 2,
  title: "Atomic Habits",
  author: "James Clear",
  category: "Self-Help",
  price: 450,
  year: 2018,
  rating: 8.5
},
{
  bookId: 3,
  title: "Wings of Fire",
  author: "A. P. J. Abdul Kalam",
```

```
  category: "Biography",
  price: 300,
  year: 1999,
  rating: 8.7
},
{
  bookId: 4,
  title: "Rich Dad Poor Dad",
  author: "Robert Kiyosaki",
  category: "Finance",
  price: 400,
  year: 1997,
  rating: 7.8
},
{
  bookId: 5,
  title: "Harry Potter",
  author: "J. K. Rowling",
  category: "Fantasy",
  price: 550,
  year: 1997,
  rating: 8.5
},
{
  bookId: 6,
```

```
  title: "The Great Gatsby",
  author: "F. Scott Fitzgerald",
  category: "Classic",
  price: 250,
  year: 1925,
  rating: 7.5
},
{
  bookId: 7,
  title: "Think and Grow Rich",
  author: "Napoleon Hill",
  category: "Self-Help",
  price: 320,
  year: 1937,
  rating: 8.0
},
{
  bookId: 8,
  title: "The Hobbit",
  author: "J. R. R. Tolkien",
  category: "Fantasy",
  price: 480,
  year: 1937,
  rating: 8.4
},
```

```
{
  bookId: 9,
  title: "Sapiens",
  author: "Yuval Noah Harari",
  category: "History",
  price: 600,
  year: 2011,
  rating: 8.2
},
{
  bookId: 10,
  title: "Ikigai",
  author: "Hector Garcia",
  category: "Self-Help",
  price: 280,
  year: 2016,
  rating: 7.8
},
{
  bookId: 11,
  title: "1984",
  author: "George Orwell",
  category: "Dystopian",
  price: 300,
  year: 1949,
```

```
  rating: 8.4
 },
 {
  bookId: 12,
  title: "To Kill a Mockingbird",
  author: "Harper Lee",
  category: "Classic",
  price: 350,
  year: 1960,
  rating: 8.3
 },
 {
  bookId: 13,
  title: "The Power of Your Subconscious Mind",
  author: "Joseph Murphy",
  category: "Self-Help",
  price: 290,
  year: 1963,
  rating: 7.9
 },
 {
  bookId: 14,
  title: "A Brief History of Time",
  author: "Stephen Hawking",
  category: "Science",
```

```
  price: 500,
  year: 1988,
  rating: 8.2
},
{
  bookId: 15,
  title: "The Diary of a Young Girl",
  author: "Anne Frank",
  category: "Biography",
  price: 320,
  year: 1947,
  rating: 8.3
},
{
  bookId: 16,
  title: "Deep Work",
  author: "Cal Newport",
  category: "Productivity",
  price: 420,
  year: 2016,
  rating: 8.1
},
{
  bookId: 17,
  title: "The Psychology of Money",
```

```
  author: "Morgan Housel",
  category: "Finance",
  price: 380,
  year: 2020,
  rating: 8.6
},
{
  bookId: 18,
  title: "Clean Code",
  author: "Robert C. Martin",
  category: "Technology",
  price: 750,
  year: 2008,
  rating: 8.4
},
{
  bookId: 19,
  title: "The Intelligent Investor",
  author: "Benjamin Graham",
  category: "Finance",
  price: 650,
  year: 1949,
  rating: 8.2
},
{
```

```
  bookId: 20,
  title: "The Fault in Our Stars",
  author: "John Green",
  category: "Romance",
  price: 330,
  year: 2012,
  rating: 7.8
},
{
  bookId: 21,
  title: "Pride and Prejudice",
  author: "Jane Austen",
  category: "Romance",
  price: 270,
  year: 1813,
  rating: 8.1
},
{
  bookId: 22,
  title: "Dune",
  author: "Frank Herbert",
  category: "Science Fiction",
  price: 520,
  year: 1965,
  rating: 8.5
```

```
},
{
  bookId: 23,
  title: "The Book Thief",
  author: "Markus Zusak",
  category: "Historical Fiction",
  price: 390,
  year: 2005,
  rating: 8.3
},
{
  bookId: 24,
  title: "Zero to One",
  author: "Peter Thiel",
  category: "Business",
  price: 360,
  year: 2014,
  rating: 7.9
},
{
  bookId: 25,
  title: "The Lean Startup",
  author: "Eric Ries",
  category: "Business",
  price: 430,
```

```
  year: 2011,
  rating: 8.0
},
{
  bookId: 26,
  title: "A Brief History of Nearly Everything",
  author: "Bill Bryson",
  category: "Science",
  price: 580,
  year: 2003,
  rating: 8.2
},
{
  bookId: 27,
  title: "The Lord of the Rings",
  author: "J. R. R. Tolkien",
  category: "Fantasy",
  price: 700,
  year: 1954,
  rating: 8.8
},
{
  bookId: 28,
  title: "The Kite Runner",
  author: "Khaled Hosseini",
```

```
  category: "Fiction",
  price: 340,
  year: 2003,
  rating: 8.4
 },
 {
  bookId: 29,
  title: "Start With Why",
  author: "Simon Sinek",
  category: "Business",
  price: 400,
  year: 2009,
  rating: 8.0
 },
 {
  bookId: 30,
  title: "The Silent Patient",
  author: "Alex Michaelides",
  category: "Thriller",
  price: 370,
  year: 2019,
  rating: 8.1
 }
])
```

```
> db.books.insertMany([
  {
    bookId: 1,
    title: "The Alchemist",
    author: "Paulo Coelho",
    category: "Fiction",
    price: 350,
    year: 1988,
    rating: 8.0
  },
  {
    bookId: 2,
    title: "Atomic Habits",
    author: "James Clear",
    category: "Self-Help",
    price: 450,
    year: 2018,
    rating: 8.5
  },
  {
    bookId: 3,
    title: "Wings of Fire",
    author: "A. P. J. Abdul Kalam",
    category: "Biography",
    price: 300,
    year: 1999,
```

Explanation: Inserts multiple books details into the collection

4. DISPLAY ALL FLIGHT

MongoDB Query:

db.books.find()

Purpose: To retrieve all 30 book records from the collection.

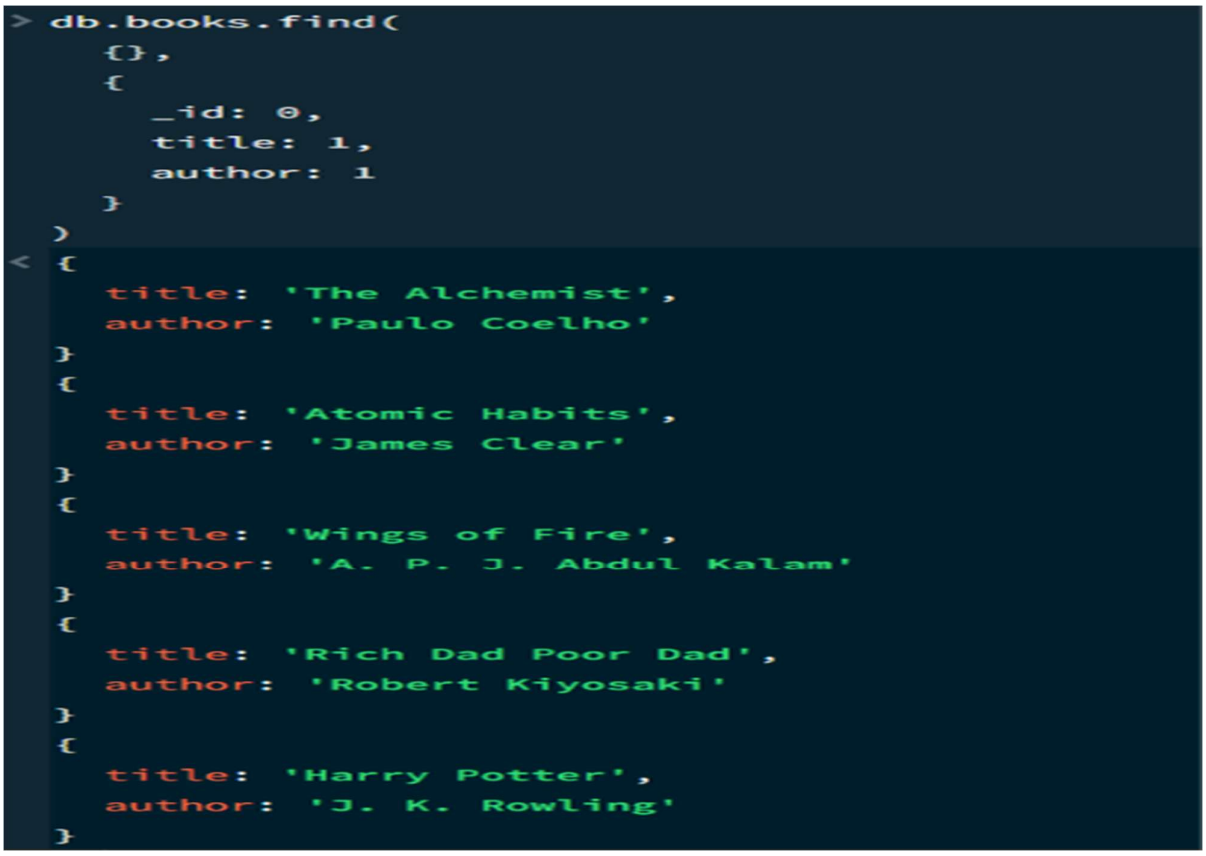
```
{
  bookId: 1,
  title: "The Alchemist",
  author: "Paulo Coelho",
  category: "Fiction",
  price: 350,
  year: 1988,
  rating: 8.0
},
{
  bookId: 2,
  title: "Atomic Habits",
  author: "James Clear",
  category: "Self-Help",
  price: 450,
  year: 2018,
  rating: 8.5
},
{
  bookId: 3,
  title: "Wings of Fire",
  author: "A. P. J. Abdul Kalam",
  category: "Biography",
  price: 300,
  year: 1999,
  rating: 8.7
```

5. Display only book titles and authors

MongoDB Query:

```
db.books.find(
  {},
  {
    _id: 0,
    title: 1,
    author: 1
  }
)
```

Purpose: To display selected fields instead of the complete document.

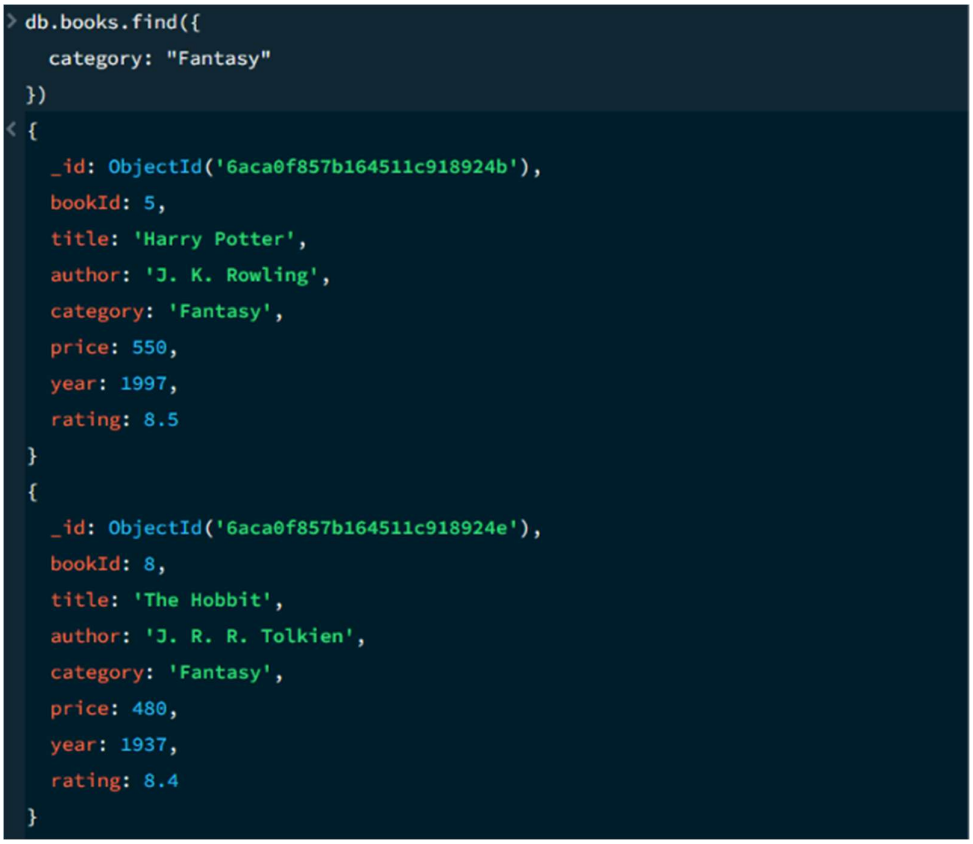


6. Find books by category

MongoDB Query:

```
db.books.find({
  category: "Fantasy"
})
```

Purpose: To find all books belonging to the Fantasy category.



7. Find books with a rating above 8

MongoDB Query:

```
db.books.find({
  rating: { $gt: 8 }
})
```

Purpose: To identify highly rated books.



8. Find books priced below ₹400

MongoDB Query:

```
db.books.find({
  price: { $lt: 400 }
})
```

Purpose: To find affordable books within a specific budget.



9. Find books published between 2000 and 2020

MongoDB Query:

```
db.books.find({
  year: {
    $gte: 2000,
    $lte: 2020
  }
})
```

Purpose: To filter books within a particular publication-year range.



```
> db.books.find({
    year: {
      $gte: 2000,
      $lte: 2020
    }
  })
< {
    _id: ObjectId('6aca0f857b164511c9189248'),
    bookId: 2,
    title: 'Atomic Habits',
    author: 'James Clear',
    category: 'Self-Help',
    price: 450,
    year: 2018,
    rating: 8.5
  }
  {
    _id: ObjectId('6aca0f857b164511c918924f'),
    bookId: 9,
    title: 'Sapiens',
    author: 'Yuval Noah Harari',
    category: 'History',
    price: 600,
    year: 2011,
    rating: 8.2
  }
```

10. Sort books by price

MongoDB Query:

```
db.books.find().sort({
  price: -1
})
```

Purpose: To display books from the most expensive to the least expensive.



11. Find the top 5 highest-rated books

MongoDB Query:

db.books.find()

.sort({ rating: -1 })

.limit(5)

Purpose: To display the five books with the highest ratings.



```
db.books.find()
  .sort({ rating: -1 })
  .limit(5)
{
  _id: ObjectId('6aca0f857b164511c9189261'),
  bookId: 27,
  title: 'The Lord of the Rings',
  author: 'J. R. R. Tolkien',
  category: 'Fantasy',
  price: 700,
  year: 1954,
  rating: 8.8
}
{
  _id: ObjectId('6aca0f857b164511c9189249'),
  bookId: 3,
  title: 'Wings of Fire',
  author: 'A. P. J. Abdul Kalam',
  category: 'Biography',
  price: 300,
  year: 1999,
  rating: 8.7
}
```

11. Update a book's price

MongoDB Query:

```
db.books.updateOne(
  { bookId: 2 },
  {
    $set: { price: 500 }
  }
)
```

Purpose: To change the price of a specific book.

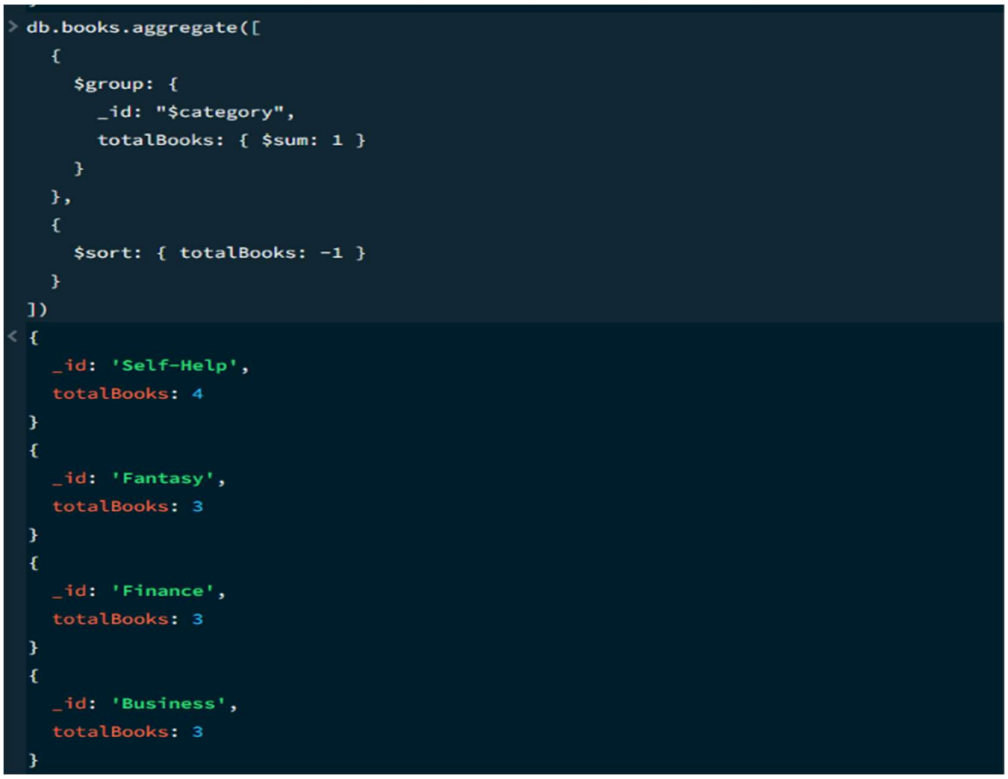


13. Count books in each category

MongoDB Query:

```
db.books.aggregate([
  {
    $group: {
      _id: "$category",
      totalBooks: { $sum: 1 }
    }
  },
  {
    $sort: { totalBooks: -1 }
  }
])
```

Purpose: To analyze the number of books available in every category using aggregation.



MONGODB OPERATION USED:

1. use — Select Database
2. createCollection() — Create Collection
3. insertMany() — Insert Multiple Records
4. find() — Retrieve Records
5. Projection — Display Selected Fields
6. Filtering — Search Records by Condition
7. Comparison Operators — $gt, $lt, $gte, $lte
8. sort() — Sort Records
9. limit() — Limit Results
10. updateOne() — Update a Record
11. aggregate() — Perform Aggregation
12. $group — Group Records by Category
13. $sum — Count Books in Each Category
14. $sort — Sort Aggregated Results

CONCLUSION

The **BookVault – Book Inventory Management System** project helped us understand the practical use of MongoDB, a NoSQL database management system. In this project, we created a database named library and a collection named books containing 30 book records.

Various MongoDB operations, such as insertMany(), find(), sort(), limit(), updateOne(), and aggregate(), were used to insert, retrieve, filter, sort, update, and analyze book information. The project also demonstrated the use of comparison operators and projection to retrieve specific book details.

In conclusion, this project provides a basic understanding of how MongoDB manages book records efficiently. It improved our knowledge of NoSQL databases and helped us learn how to perform different database operations using MongoDB queries.